The Bilinear GCIQL Critic Is Not a Learned Distance Function

Evidence from OGBench cube manipulation, and a mathematical account of what the critic learns instead

August 2026

Setting. All experiments are offline goal-conditioned RL on OGBench `cube-single-play-v0` (state-based single-cube manipulation, one million transitions of scripted, task-agnostic play data), 3 seeds × 500,000 gradient steps per method. The policy is a flow-matching behavior-cloning actor plus a critic-value ascent term; a control actor with the critic term removed isolates the critic's contribution. Methods compared: goal-conditioned IQL with a standard MLP critic (**GCIQL**), the same objective with a bilinear critic (**bilinear GCIQL**), contrastive RL (**CRL**), and quasimetric RL (**QRL**) as a reference that enforces distance structure by architecture.

1 The hypothesis under test

The bilinear critic is $\hat{Q}(s,a,g) = \langle \varphi(s,a), \psi(g) \rangle / \sqrt{d}$ with value $\hat{V}(s,g) = \langle \varphi(s), \psi(g) \rangle / \sqrt{d}$, trained by goal-conditioned expectile TD (expectile 0.9, discount $\gamma = 0.99$) on goal-relabeled data. Let $d^*(s,g)$ be the minimal number of steps from s to g supported by the dataset. Any value function induces an estimated distance $\hat{d}(s,g) = \log_\gamma\big(1 + (1-\gamma)\hat{V}\big)$, a monotone transform of $\hat{V}$; every measurement below is rank-based and therefore identical for $\hat{V}$ and $\hat{d}$.

Hypothesis 1 (The distance hypothesis)**.** *Bilinear GCIQL outperforms GCIQL (*92.7 ± 1.4 *vs.* 73.2 ± 1.3 *success rate; the behavior-cloning control reaches* 8–10*) because its* $\hat{d}$ *approximates the true distance* d^* *more faithfully, and this faithfulness is what improves the policy.*

If the hypothesis holds, the bilinear critic must (a) rank pairs by temporal distance at least as well as GCIQL on held-out data, (b) agree with the methods that estimate distances well by construction (QRL, CRL), and (c) improve its distance estimates over the training window in which its success rate improves.

2 The evidence: all three implications fail

Finding 1 (GCIQL estimates temporal distance better, and loses)**.** *Spearman rank correlation between* $\hat{d}(s,g)$ *and the time separation of* (s,g)*, over* 10^5 *held-out same-trajectory pairs (second number: partial correlation controlling for Euclidean distance between raw observations):*

$$\textit{GCIQL}:\ 0.63\,/\,0.54 \qquad > \qquad \textit{bilinear GCIQL}:\ 0.54\,/\,0.41.$$

The critic with the worse *distance estimate wins by 20 success points. Distance faithfulness on the data distribution is therefore neither sufficient (GCIQL) nor necessary (bilinear GCIQL) for policy improvement.*

Finding 2 (The bilinear critic disagrees with every good distance estimator)**.** *Compute each method's* 2000×2000 *matrix of pairwise estimated distances on a fixed state sample, and correlate the matrices between methods (Spearman on the upper triangles). GCIQL–QRL correlate at* 0.88 *and GCIQL–CRL at* 0.54*: the three distance-flavored methods agree with one another. The bilinear critic agrees with none of them (at most* 0.21*) and is instead closest to plain Euclidean distance between raw observations (*0.45–0.52*). If it had learned the temporal distance, it would agree with the methods that demonstrably do; instead, those methods underperform it by 20–60 success points.*

Finding 3 (The similarity structure is an artifact of the architecture, not of learning)**.** *After training, the bilinear output is numerically a scaled cosine similarity: the correlation between* $\langle \varphi, \psi \rangle$ *and* $\cos \angle(\varphi, \psi)$ *is* 0.996*, and the embedding norms vary by under* 9% *across inputs. But the* untrained *network already shows correlation* 0.994*: layer normalization concentrates the norms at initialization. "It learned a cosine similarity" is a property of the architecture and cannot explain an advantage acquired during training.*

Finding 4 (Distance quality plateaus while performance keeps rising)**.** *Between 100k and 500k gradient steps, the rank correlation of Finding 1 moves only* $0.37 \to 0.41$ *while success moves* $65 \to 93$*. The quantity the hypothesis says is causal is flat over the window in which the effect it must explain triples.*

Finding 5 (Where the two critics really differ: states visited at evaluation time)**.** *Roll out each trained policy on the evaluation tasks (deterministic actions, 3 seeds* $\times$ *50 episodes) and record, for each episode, the Spearman correlation between* $\hat{V}(s_t, g)$ *and the timestep* t *— for a sound value function, value should rise as the policy approaches its goal. Score* both *critics on* both *policies' trajectories (median per episode, mean*$\pm$*sd over seeds):*

	trajectories of the bilinear-GCIQL policy (success 0.95*)*	*trajectories of the GCIQL policy (success* 0.73*)*
bilinear-GCIQL critic	0.983 ± 0.003	0.871 ± 0.060
GCIQL critic	$\mathbf{0.355 \pm 0.239}$	0.883 ± 0.015

On held-out dataset pairs GCIQL ranks better (Finding 1); on the states actually visited by the successful *policy at evaluation time, its value estimates are close to uninformative (on one task the median correlation is* -0.80*: value falls as the policy approaches the goal), while the bilinear critic's value rises almost perfectly monotonically. Each critic is coherent along its own policy's trajectories (diagonal); only the bilinear critic remains coherent on the* other *policy's states. What separates the two critics is not distance quality on the data distribution but coherence on state–goal pairs* outside *it — exactly the pairs evaluation interrogates.*

Remark 1 (Verdict)**.** *Implication (a) fails by Finding 1, (b) by Findings 2–3, (c) by Finding 4, and Finding 5 shows the real discriminator lies elsewhere. The bilinear critic also freely violates the defining axioms of a distance where it wins — its estimates are asymmetric and violate the triangle inequality at rates comparable to GCIQL's. Whatever it learns, calling it a distance function explains none of the observed orderings.*

3 What the critic is plausibly learning instead

The surviving account: the bilinear critic learns a **bounded, low-rank approximation of the optimal goal-conditioned value function**, and its advantage comes from how that function class behaves under policy optimization — not from any distance structure. The argument has three steps, each tied to a measured quantity.

Setup. With reward -1 per step until the goal is reached, the optimal value is $V^*(s, g) = \big(\gamma^{d^*(s,g)} - 1\big)/(1-\gamma) \in [-100, 0]$ at $\gamma = 0.99$. Expectile regression with expectile near 1 pushes $\hat{V}$ toward this in-sample optimum on the distribution μ of relabeled state–goal pairs. Because layer normalization concentrates $\|\varphi\|$ and $\|\psi\|$, the bilinear class is uniformly bounded:

$$|\hat{V}|, |\hat{Q}| \;\le\; \kappa := \|\varphi\|\,\|\psi\|/\sqrt{d}, \qquad \kappa \approx 75 \text{ as measured, realized } \hat{V} \in [-63, +6].$$

Step 1: the bound caps how much the actor can exploit critic error. The actor maximizes $\hat{Q}$ against a behavior-cloning anchor. For any policy π the standard decomposition gives

$$J(\pi) \;\ge\; J(\pi_{\mathrm{BC}}) \;+\; \mathbb{E}_{s,g}\big[\hat{Q}(s, \pi) - \hat{Q}(s, \pi_{\mathrm{BC}})\big] \;-\; 2 \sup_{s,a,g} \big|\hat{Q} - Q^*\big|.$$

For an unbounded MLP the supremum is uncontrolled precisely outside the data distribution, where the maximization goes hunting; in the bilinear class it is at most $\kappa + \sup|Q^*|$. Indeed $\kappa \approx 75 < 100 = \sup|V^*|$:

the class cannot even represent the most extreme values, let alone hallucinate beyond them. Errors are forced to be many, small, and spread out rather than rare and enormous — matching the measurement that GCIQL's triangle-inequality violations average 9.5 steps in magnitude against the bilinear critic's 4.8 at comparable rates.

Step 2: low rank forces the critic to fill in unseen state–goal pairs consistently. View the value function on the dataset's states and goals as a matrix $M_{sg} = \hat{V}(s, g)$; the bilinear form constrains $\text{rank}(M) \leq d$. Goal relabeling covers each state with *some* goals and each goal with *some* states, but not all combinations; evaluation asks about combinations never seen together. A rank-d matrix has only $d(|\mathcal{S}|+|\mathcal{G}|-d)$ degrees of freedom and — this is the standard matrix-completion fact — is uniquely determined by a number of generically-positioned entries proportional to $d(|\mathcal{S}| + |\mathcal{G}|)$, up to logarithmic factors. Once training pins down the two factors on the covered pairs, the values on the *uncovered* pairs are no longer free: the class extrapolates them consistently. An MLP critic obeys no such constraint; its values off the training distribution are limited only by smoothness. This is Finding 5 in mathematical form: the bilinear critic stays coherent on another policy's states (0.87) where the MLP critic collapses (0.36).

Step 3: what is learned is a low-rank approximation of V^*, not a distance. In the idealized limit, expectile TD in the rank-constrained class solves

$$\min_{\text{rank } M \leq d} \left\| W \odot (M - V^*) \right\|_\mu^2,$$

a weighted low-rank approximation of the optimal value function — i.e. the critic retains (a reweighting of) the top-d singular subspace of V^*. This predicts, and the data confirm:

(i) performance saturates once d exceeds the effective dimension of that subspace: measured success 0.06/0.62/0.92/0.90/0.93 at $d = 5/10/20/30/50$, saturating between $d = 10$ and $d = 20$, while the trained embeddings' effective dimension keeps growing with d (4.1/8.0/14.7/19.6/28.0) — capacity, not a learned constant;

(ii) the axioms of a distance (symmetry, triangle inequality) are incidental — a truncated low-rank approximation of V^* satisfies neither exactly, and the bilinear critic indeed violates both while winning;

(iii) distance measurements on the data distribution (Findings 1, 2) can prefer GCIQL while the actor prefers the bilinear critic, because such measurements are blind to both the bound of Step 1 and the off-distribution consistency of Step 2.

Proposition 1 (Summary of the account, and how it can fail)**.** *The bilinear critic's advantage = a uniformly bounded ($\kappa < \sup|V^*|$), rank-d approximation of the optimal goal-conditioned value function, extended to unseen state–goal pairs by the low-rank constraint. Falsifiable consequences, currently under test: (a) imposing the same bound κ on the MLP critic (squashing its output through $\kappa \tanh(\cdot/\kappa)$) should recover part of the $73 \to 93$ gap; (b) removing the layer normalization from the goal embedding, which destroys the uniform bound at fixed rank, should degrade the bilinear critic; (c) the off-distribution coherence of Finding 5 should vanish for d below the saturation dimension. Any of these failing narrows the account; none of them restores Hypothesis 1, which Findings 1–5 refute independently.*